

Measurement of soil macropore geometry by image analysis of sections through impregnated soil

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Received 5 September 1995. Accepted in revised form 31 January 1996

Key words: image analysis, pattern recognition, soil pore space, soil structure, stereology

Abstract

The geometry of the soil pore space affects root growth directly by providing pathways for root extension and indirectly through its effects on soil aeration and on water infiltration, redistribution and drainage. Image analysis of sections through impregnated blocks of undisturbed soil allows quantitative assessment of the pore space. Samples are dehydrated and impregnated with resin containing a fluorescent dye. Once the resin has hardened, the blocks can be cut to reveal a section through the structure. An image of the pore space, which is now filled with resin, can be obtained using ultra-violet light either photographically or digitally. Digital images are segmented into pore space and solid. Image analysis techniques can be used to classify the pore space into channels, fissures and packing pores. This allows appropriate measurements to be made on each class so that stereology can be used to estimate 3-D parameters from the measurements made on the image. Various indices can also be derived to quantify the pore structure.

Introduction

Efficient exploitation of the soil by roots depends on adequate root extension, which can be hindered by many soil factors including chemical toxicity, nutrient deficiency, anaerobic conditions or lack of water. In the absence of these factors, root extension will be a function of soil strength and soil structure. Roots are able to extend in several ways. If soil strength is sufficiently low, they can penetrate the soil matrix directly. Alternatively, they can use the macropore space as a means of extending either by following macropores of greater widths than themselves or by widening smaller macropores using their internal pressure.

Various types of penetrometers have been used to measure soil strength, which is itself a function of the strength of individual aggregates or structural units (peds) and the structure. Penetrometers, in particular micro-penetrometers which are closer in size to roots, can give an indication of the resistance encountered by a growing root. However, being rigid, they cannot

simulate the ability of roots to grow in soil of high strength by following macropores.

The extent, shape, size and connectivity of the macropore space all affect how it can be exploited by growing roots. These factors also affect many physical processes which influence root growth, including water infiltration into, movement through and storage in the soil profile together with gaseous movement. It is possible to produce images of the macropore space by taking sections through undisturbed soil which has been impregnated with resin. These images can be digitised and measured using image analysis.

In the 1970's, the first image analysers became available and their potential for analysis of soil structure was realised by Jongerius et al. (1972) and Foster and Evans (1971). Jongerius et al. (1972) and Ismail (1975) used the Quantimet B and Quantimet 720 (Cambridge Instruments), respectively. They were mainly concerned with simple measurements of pore size. Murphy et al. (1977a), Bouma et al. (1977), Bullock and Murphy (1980) and Ringrose-Voase and Bullock (1984) used the Quantimet 720 to analyse pore shape in various ways. Ringrose-Voase (1987) used shape anal-

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ysis of pores to develop quantitative descriptions of soil morphology. McBratney and Moran (1990) measured depth functions for various structural attributes which gave information about changes of structure with depth at very small depth intervals.

Dexter (1976), Dexter and Hewitt (1978) and Hewitt and Dexter (1981) developed a method of deriving some structural indices from transects across images of soil pore structure. Ringrose-Voase (1990) describes some 1-D parameters measured by image analysis which can be used as structural indices.

There are now available a variety of purpose-built image analysers and image analysis software for personal computers and workstations. Image analysis has been used to study the effects on soil structure of a variety of soil treatments. These include compaction (Murphy et al., 1977b); cultivation system (Dexter et al., 1982, 1983; Hewitt and Dexter, 1980; McKenzie et al., 1994; Moran et al., 1988; Pagliai et al., 1983, 1984, 1989a; Shipitalo and Protz, 1987); puddling of rice soils (Pagliai et al., 1987b); cropping systems (McKeague et al., 1987); organic matter (Pagliai et al., 1987a) and artificial organic additives (Pagliai et al., 1980); gypsum (Chartres et al., 1985; O'Callaghan and Loveday, 1973); depth to water table (Pagliai et al., 1989b) and soil drying (Guidi et al., 1978). Soil loss as affected by crusts and surface seals were studied using image analysis by Norton (1987), Norton and Schroeder (1987) and Carnicelli et al. (1994). The orientation of pore space in Vertisols was investigated using image analysis by Bui and Mermut (1989) and Hallaire (1994). Bui et al. (1989) and Curmi et al. (1994) used image analysis to investigate the bimodal size distribution of pore sizes in Oxisols. Bullock and Thomasson (1979) and Vogel and Babel (1994) compared pore size distributions derived from image analysis and from water retention measurements and D'Acqui et al. (1994) did likewise for mercury porosimetry measurements. Walker and Trudgill (1983) and Nortcliff et al. (1994) compared pore geometry as measured by image analysis with that derived from tracer breakthrough curves. Hallaire and Curmi (1994) measured pores taking part in water movement. Bouma et al. (1977, 1979) and Bouma and Wösten (1984) used structural data from image analysis to model water movement in structured clay soils. McKeague and Fox (1987) used image analysis to check the accuracy of field estimates of porosity, while Ringrose-Voase (1987) and Ringrose-Voase and Nys (1990) compared image analysis parameters with field descriptions of morphology.

Stewart et al. (1994) and Krebs et al. (1994) have made advances in analysing the distributions of roots in relation to the pore space.

Measurements of the macropore space made by image analysis are one- or two-dimensional. However, they can be used to estimate three-dimensional parameters using stereology provided a suitable geometric model is available (Ringrose-Voase and Nortcliff, 1987). In cases where there is no such model, 2-D measurements can still provide useful indices of the macropore space which could be related to root growth.

This paper is concerned with image analysis of macropore space as seen in sections through intact soil samples which have been impregnated with resin. However, some of the image analysis methods described are also useful for images obtained by alternative means. For example, images of smaller pores can be prepared from sections through impregnated samples using scanning electron microscopy in back-scattered mode (Bisdorf and Thiel, 1981). 2-D and 3-D images of the pore space can be obtained non-destructively using computed tomography (CT). CT is beyond the scope of this paper and has been reviewed by Aylmore (1993), Heijs et al. (1995) and various workers in Anderson and Hopmans (1994).

Sample preparation

Sampling

The first and most crucial step in producing images of soil macropore space is obtaining samples in which the structure is as undisturbed as possible. There are several ways of doing this which generally involve inserting a container with two open faces into a ledge excavated at the desired depth in the soil. Rectangular tins are inserted horizontally or vertically by cutting away the soil outside the tin with a knife and then pushing the tin. This is repeated in small steps until the tin is fully inserted (e.g. Murphy, 1986). Typical tin sizes are 75 × 65 mm × 40 mm deep; 205 × 125 mm × 40 mm deep and 160 × 120 mm × 80 mm deep. Alternatively, cylindrical cores can be taken using aluminium or steel rings with one end sharpened. Brass should not be used because it inhibits polymerisation of some resins. Typical sizes are 50 mm diameter × 60 mm depth; 100 mm diameter × 75 mm depth and 110 mm diameter × 105 mm depth. Such rings are inserted vertically using a drop hammer, explosive charge or

by jacking against a vehicle (McIntyre, 1974a, b). A guide ring placed on the soil surface should be used to ensure that insertion is vertical. The sharpened leading edge minimises compression inside the sample. After sampling, the exposed faces of the sample should be covered with lids to prevent damage and the sample should be wrapped in plastic to prevent water loss.

Compared to rectangular tins, cores have the advantages of being easier to take and of being approximately equidimensional. The latter means that after the core has been cut and photographed at one orientation the same can be done at another orientation by realigning the two halves and gluing them together. They have the disadvantage that in soils with low macroporosity they are more difficult to impregnate because of the longer distance to the centre of the core.

It is also possible to impregnate soils in the field. This is particularly useful where the structure is too loose to be sampled by normal methods without disruption. A gypsum slurry can be poured on to the surface and allowed to infiltrate into the macropore space (FitzPatrick et al., 1985; Kohno, 1987). After the gypsum has hardened, the soil is excavated to reveal the pattern of macropores which can be photographed or drawn. The gypsum-impregnated soil can also be sampled and further impregnated with resin in the laboratory (FitzPatrick et al., 1985). Moran et al. (1989a) developed a method of field impregnation with resin.

Whatever the method of sampling, replicate samples must be taken to allow for field variability. Murphy and Banfield (1978) recommend that six 75 × 60 mm samples be taken for each horizon to be investigated. However, the number required will depend on the size of the samples and the variability of the parameters being measured. The relative standard error of some parameters is discussed below.

Dehydration

Before samples can be impregnated with resin they must be dehydrated. Soils with low clay content can be air- or oven-dried without changing the structure due to shrinkage. In soils with appreciable clay contents, water can be removed by freeze-drying (Stephan, 1969; Jongerius and Heintzberger, 1975) or solvent replacement in the vapour phase (FitzPatrick, 1974) or liquid phase (Miedema et al., 1974). Murphy (1982) found the latter to be most suitable. The most commonly used solvent is acetone but Greene-Kelly (1973) found that methanol or dioxane caused less shrinkage. Chartres et al. (1989) recommend dioxane for smectitic

soils in particular, since acetone can cause samples of such soils to collapse.

Replacement in the liquid phase involves immersing samples in solvent until solvent and water are mixed, after which the mixture is replaced. This continues at approximately weekly intervals until most of the water has been removed. Between six and fifteen weeks are required, depending on sample size and water content. Murphy (1985) found that equilibration was faster as dehydration progresses so that the total time required could be decreased by using progressively shorter intervals. Moran et al. (1989b) developed a system of tanks which hold up to 64 samples. To hasten equilibration, solvent is circulated through the tanks and is temporarily drained each day into a reservoir to empty the macropore space. As the solvent circulates, it is itself dehydrated using a water-absorbing polyacrylamide polymer and by a type 4 Å molecular sieve when low water contents are reached. The water content of the solvent mixture is monitored enthalpimetrically.

Impregnation

When samples have been dehydrated they are removed from the solvent and allowed to drain. However, it is important that they remain wet with solvent otherwise shrinkage will occur (Chartres et al., 1989). Samples are placed in containers to which resin mixture is added ensuring that the samples are not covered so that air and solvent can escape. Clear, polyester, casting resin is the most widely used impregnant. Epoxy resins can also be used. Monomer is added to make the mixture less viscous and ultraviolet dye is included to make it fluorescent. The samples are left for 24 hours or longer to allow the mixture to be taken up by capillary action. They can then be evacuated for up to 24 hours to remove any remaining air and solvent before being covered with mixture and left to harden. The rate of hardening is controlled by the amount of catalyst, which should be such that hardening takes about six weeks. If hardening is too rapid, heat will be generated and the samples become distorted.

The technique of Moran et al. (1989a) for impregnating soil in the field uses a reactive diluent to reduce the viscosity of the resin mixture, which is capable of hardening in 24 hours even in moist soil. The impregnated volume is excavated from the soil when hard. In the laboratory, it is cut and reimpregnated to fill any empty pore space before being polished and pho-

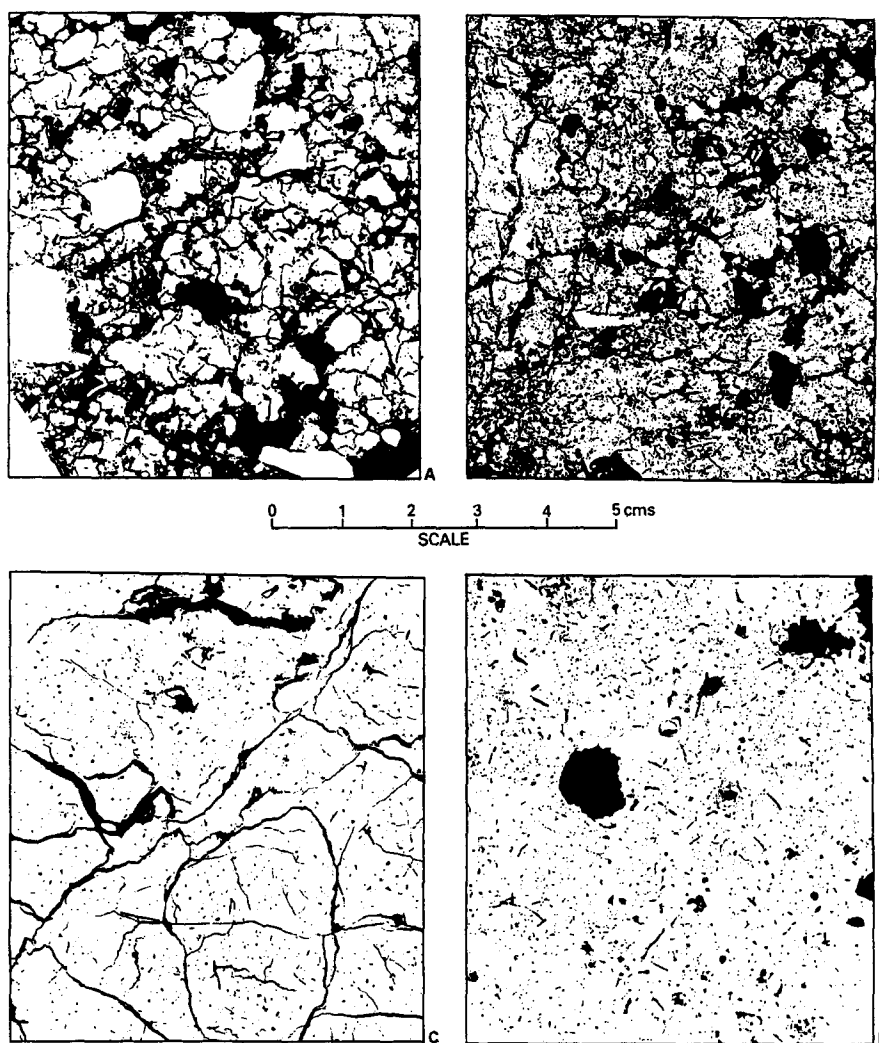


Figure 1. Examples of images of soil macropore space. (From Ringrose-Voase and Bullock, 1984).

tographed. This technique is particularly useful in friable topsoils or where results are required quickly.

Cutting and polishing

After hardening, samples are cut using a diamond-edged saw. One of the cut faces is polished using abrasive Carborundum paper followed by a lapping wheel with an abrasive slurry. Murphy (1986) gives some alternative techniques.

Fluorescent photography

If the face revealed by cutting the sample is illuminated with ultraviolet (UV) light, the resin fluoresces. Since resin occupies the pore space, the pattern of

fluorescence is a 2-D image of the macropore space and can be recorded photographically. The method described here is similar to that of Geyger and Beckmann (1967) as modified by Ringrose-Voase (1984). The sample is placed facing a camera (in a dark room) in a specially-constructed stand which holds the face to be photographed parallel to the film. Either 35 mm or 4 × 5 inch (102 × 127 mm) format cameras are suitable. Two UV lamps (100 W each) are mounted above the block. The lamps should have visible light filters so that only UV light reaches the sample. In addition, the camera lens should have a UV filter so that only visible light produced by fluorescence exposes the film. If high contrast film is used, the resultant negatives clearly show the macropore space (black) and the soil matrix (white) as in Figure 1.

Generation of digital images of macropore space

Photographs of soil macropore space can provide useful information in a qualitative sense, but this information is often subjective and cannot be subjected to statistical analysis to compare samples from different soils or treatments. Photographs can be analysed manually using point counting (e.g. Murphy, 1983) or test lines (e.g. Scott et al., 1986), but such methods are time-consuming. Computerised image analysis offers a means of rapidly analysing images.

Digitising

Digital images of soil macropore space can be obtained directly from samples illuminated by UV light in a manner similar to that described for fluorescent photography but using a video or CCD (charged couple device) camera. Alternatively, photographs of the pore space can be digitised using a video or CCD camera or flatbed scanner. The digitised image consists of a 2-D matrix of picture elements or pixels which cover the area of the image. The size of each pixel depends on the magnification of the lens used but is commonly between 10 and 200 μm . The number of pixels per image should be at least 500×500 . Each pixel has a grey-level value which depends on its brightness and is usually between 0 and 255. Pixels can be arranged on square or hexagonal rasters. Hence, if the images in Figure 1 were digitised, pixels representing macropore space (black) would have small grey-level values and those representing the soil matrix (white) large values.

The advent of colour scanners gives the possibility of generating multi-channel digital images of thin sections or polished blocks which would enable analysis of other soil components in addition to pore space. Each channel or layer in such images is a view of the same scene but the pixel values represent the brightness of a different colour. Terribile and FitzPatrick (1992) generated multi-channel images of soil thin-sections using optical microscopy with a variety of illumination methods (plain transmitted light, crossed polarisers, circularly polarised light and circularly polarised light with a gypsum plate). Each illumination method was viewed using red, green and blue filters or no filter. They selected a subset of the 16 channels obtained that best distinguished quartz, clay coatings and matrix in addition to pore space.

Image segmentation

The next step is to *segment* the digital image of grey-level values into a binary image showing macropore space labelled as 1 and solid as 0. The simplest procedure is to choose a threshold value somewhere between the low values representing pore space and the high ones representing matrix. However, although the histogram of pixel frequency for each brightness value usually shows a bimodal distribution, there is a considerable proportion of pixels with intermediate values. This is caused by variation in illumination across the image and because the value of each pixel is the brightness integrated over a finite area. A large proportion of intermediate pixels means that a small change in threshold value may cause considerable changes to the number of pixels labelled as pore space.

There are various alternative segmentation techniques which overcome the problems of thresholding. One such method has the following stages:

1. Reverse the image by subtracting it from 255 (i.e. make dark pixels bright and vice versa).
2. Choose two threshold values from the histogram of the pixel values. All pixels greater than the upper threshold (P) should be clearly macropore space and all less than the lower one (R) should clearly be matrix. Pixels between the two threshold values are the grey pixels around the edges of the macropore space.
3. The digital image is subjected to edge enhancement, in this case using a *Laplacian filter*. The filter consists of a structuring element which is multiplied by the neighbourhood of each pixel in the input image to produce a second image. For example, the structuring element to apply a Laplacian filter and add the result to the original image is (for an hexagonal raster):

$$\begin{array}{ccc} & -1 & -1 \\ -1 & & 7 & -1 \\ & -1 & -1 \end{array}$$

The value of the output pixel consists of the sum of the input pixel and its six neighbours, each multiplied by the relevant factor in the structuring element. If the input pixel is surrounded by pixels of the same grey-level value, the output pixel has the same value. If the neighbourhood pixels have different values (i.e. near an edge), the output pixel is increased or decreased in value depending on the gradient of the change. This has the effect of sharpening edges.

4. The image from (3) is segmented in two ways.
 - a) A threshold value of P is applied which sets all pixels with value equal to or greater than P to 1 and all others to 0. This binary image shows everything that is definitely pore space as 1. However, narrow regions of the pore space are often missing. These include necks between wider regions of pore space, which may be related to the 3-D connectivity.
 - b) A threshold value of R is applied separately to the image from (3) to create a second binary image in which pixels with values greater than R are labelled 1. These are either definitely pore space or part of the grey area. This image shows the pore space to be thicker than in 4a. However, the necks and other narrow regions are usually present.
5. A morphological thinning is applied once to image 4b. This is a logical operation using a structuring element such as:

$$\begin{array}{ccc} & 1 & 1 \\ - & 1 & - \\ & 0 & 0 \end{array}$$

The element or any of its rotations is compared to each pixel and its neighbourhood in the image. If it matches the neighbourhood ("-" indicates either 0 or 1) the output pixel is set to 0. Hence, pore pixels (1) which are adjacent to matrix pixels (0) are set to 0, unless the pore space at that point is only one pixel thick. In the resulting image, the pore space has been thinned (often becoming narrower than in image 4a) but narrow regions have been retained.

6. Finally, images 4a and 5 are combined using a logical OR operator to create an output image in which pixels are set to 1 if they are 1 in either image 4a or 5. This allows the necks and other narrow regions retained in image 5 to be combined with the main parts of the pore space in image 4a.

Moran et al. (1988, 1989a) describe two similar methods for segmenting images. Multi-channel images produced by digitising in colour or using different illumination methods such as those of Terribile and Fitz-Patrick (1992) can be segmented using image processing methods normally used for multi-channel, remotely sensed images. The statistics of each component in terms of the brightness of its pixels in each of the channels is determined by selecting training areas for each component. These statistics are then used to classify all pixels in the image.

Image measurements

The segmented image, showing macropore space and solid, is used for measurement. Parameters can broadly be divided into two types: image parameters, which refer to the whole image, and object parameters, which refer to discrete objects within the image.

Image parameters

Area density, A_A (pore area / unit area of image), is measured by counting the number of pore pixels and dividing by the total number of pixels. The length of pore perimeter per unit area is known as the perimeter density, B_A . The length density, L_A , for example of a line produced by reducing the pore space in the image to a skeleton, is defined similarly.

Numerical density, N_A , (number of objects/unit area) applies to the number of discrete objects intersected by the 2-D plane of the image. When counting objects or intercepts, steps must be taken to avoid bias due to the edge effect, as discussed below with object parameters.

The above parameters suffixed by subscript A are 2-D parameters. However, it is also possible to measure 1-D parameters by treating the scan lines as test lines through the image. Such parameters are suffixed by subscript L. The intercept density, I_L , is the number of intercepts the test line makes with pore edges per unit length of test line. On the other hand, N_L is the number of intercepts/unit length with pore space. Usually N_L is equal to approximately $I_L/2$. L_L is the length of test line intercepting pore space per unit length.

Object parameters

Objects are discrete groups of adjacent pixels set to 1 and completely surrounded by pixels set to 0. Before objects can be measured, they must first be delimited by assigning each pixel within an object the same object label. Object parameters include the area, A , perimeter, Pe , and calliper diameters, H , of individual objects. Complex parameters can be calculated from these. For example, convex perimeter, $ConPe$, can be calculated from the calliper diameters:

$$ConPe = \frac{\pi}{2} \cdot \sum_{i=1}^N H_i \quad (1)$$

where N is the number of calliper diameters measured.

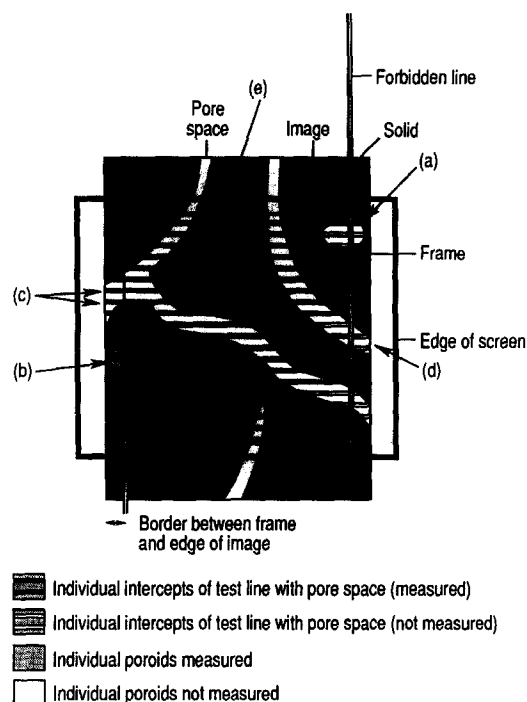


Figure 2. Use of a border area and forbidden line to decide which poroids or intercepts to measure. When measuring individual intercept lengths: a) those whose right end is outside the frame are not measured; b) those whose left end is outside the frame are measured in their entirety and c) those crossing the edge of the image will be truncated if the edge is in the border region. When measuring individual poroids: d) those touching the forbidden line are not measured and e) the remainder are measured in their entirety. When measuring total pore area, perimeter etc., all pore space within the frame is measured and none outside. (After Ringrose-Voase, 1990).

Steps must be taken to avoid bias due to the edge effect when deciding which objects to count or measure. A reference frame is created within the image which leaves a border area above and to the left of the image (Fig. 2). The width of the border should be larger than most commonly occurring objects in the image. The rule to avoid bias is to include only objects in the frame which do not touch a "forbidden line" (Gundersen, 1977) running along the bottom and right edges of the image (see Fig. 2). Objects which have a part within the reference frame but which project into the border area should be measured in their entirety, unless they touch the forbidden line.

When images are analysed in 1-D using test lines, the only "objects" are the intercepts between test line and macropore space. These intercepts have a single parameter, namely length. When counting or measuring individual intercepts the same edge rules apply as when measuring 2-D objects. An intercept is measured

in its entirety if part of it falls within the reference frame unless it crosses the forbidden line, in which case it is not measured.

Recognition of pore types

Types of soil pore space

Soil macropore space is formed by various processes resulting in different shapes. Brewer (1964) recognised four major pore types: packing pores, vughs, channels and planes (referred to here as fissures). Different types of macropore space have different physical properties with respect to water and gas movement and in their interactions with growing roots. Because pore types have different shapes in 2-D, image analysis can be used to allocate pore space in an image to pore classes and provide quantitative information about each class. This information can be related to macropore forming processes and to aspects of soil behaviour which depend on macropore structure. Separation of the macropore space into shape classes allows appropriate geometrical models to be applied to each class. Such models allow stereology (see below) to be used to estimate 3-D parameters from the 2-D or 1-D measurements (Ringrose-Voase and Nortcliff, 1987).

The macropore space in each of the images in Figure 1 is dominated by one of three pore types. Figures 1a and b are dominated by packing pores, Figure 1c by fissures and Figure 1d by channels.

Packing pores result from the loose packing of single grains, aggregates or peds. They have irregular shapes in both 3-D and 2-D. They are highly interconnected in 3-D and often appear so in 2-D when sectioned. Ringrose-Voase (1987) considers them structural pores since they define structural units (peds and aggregates).

Vughs were considered by Ringrose-Voase and Bullock (1984) to be packing pores which have become disconnected through settling or compaction. Hence, vughs can be placed in the packing pore class because the difference between them is one of connectivity rather than shape.

Channels are created by both roots and soil fauna. They are essentially cylindrical but may be tortuous. When sectioned, they appear in 2-D as circles if cut normal to their axis or as ellipses if cut obliquely. Channels are trans-structural because they do not define structural units and may pass through or between them (Ringrose-Voase, 1987).

Fissures arise from shrink-swell processes and other mechanical stresses. They are curved, sheet-like structures of varying width, which is negligible compared to the extent of the sheet. When sectioned they appear as long lines of varying width, which taper to sharp ends. They may be more or less connected depending, for example, on the degree of shrinkage. Like packing pores, they are structural pores since they define structural units (Ringrose-Voase, 1987).

Statistical pattern recognition

Ringrose-Voase and Bullock (1984) developed a system, ANOPOR, which allowed individual poroids¹ in an image to be allocated to one of three classes, packing pores, channels or fissures, on a statistical basis. It is difficult to define boundaries between each class both because there is a variety of shapes within each class and because intergrades exist. The statistical method described allows the classes to be defined by selecting representative examples of each class, which make up a learning set. Poroids in an image are compared statistically to each pore in each class of the learning set and the probability of belonging to each class is calculated.

Learning set

A learning set was created by selecting representative poroids of the three classes from a collection of over 10,000 samples. These were isolated from images of the samples to create the set shown in Figure 3.

Shape factors

Each poroid was described by shape factors S (shape) and C (convex shape), which increase as poroids deviate from a circle:

$$S = \frac{Pe}{A^{\frac{1}{2}}} \quad (2)$$

$$C = \frac{ConPe}{A^{\frac{1}{2}}} \quad (3)$$

where $D \geq 1$. Two kinds of deviation are measured. Small scale irregularities in the perimeter will increase S but leave C unaffected. Larger indentations or elongation will increase both S and C . The magnitude of $Pe/A^{1/2}$ is more limited as poroid area decreases because pixel size limits the resolution of irregularities along poroid perimeters. D corrects for this in the calculation of S .

¹ Defined by Moran et al. (1988) as the individual, discrete outlines seen in 2-D which result from sectioning the pore space.

If the learning set is plotted in the pattern space of S and C , the data points show there is correlation between the two variables and a clustering of points at low values. Both these factors limit the usefulness of S and C as descriptive variables. Instead the learning set was plotted in a transformed pattern space of r and θ (Fig. 4) where:

$$r = \sqrt{S^2 + C^2} \quad (4)$$

$$\theta = \arctan \left(\frac{S}{C} \right) \quad (5)$$

In the transformed pattern space r and θ are found to be more independent where r is a measure of poroid elongation and θ is a measure of irregularity.

The value of D was calculated iteratively by maximising inter-class separation and intra-class clustering in the r versus θ pattern space. A measure of this for the poroids in the learning set in one class (ω_p) relative to those in another (ω_q) is the ratio, $R(p,q)$, of the mean inter-class probability density to the mean intra-class probability density. The probability density of one poroid in class ω_p relative to those in another ω_q is defined later in Equation 8.

$$R(p,q) =$$

$$\frac{\text{Mean probability density of } \omega_p \text{ with respect to } \omega_q}{\text{Mean probability density of } \omega_p \text{ with respect to } \omega_p} \quad (6)$$

The ratio for all classes is $R(P,C) \cdot R(C,P) \cdot R(P,F) \cdot R(F,P) \cdot R(C,F) \cdot R(F,C)$, where P is the packing pore class, C the channel class and F the fissure class. The optimum value of D was selected as 1.178 by minimising this ratio over a range of D values.

Poroid allocation

Unknown poroids are allocated to a class by comparing them to those in the learning set, using a non-parametric measure of similarity to calculate the probability of belonging to each class (Coomans and Marsart, 1981). The position of an unknown poroid (i) in an R -dimensional pattern space is described by its pattern vector, $\vec{x}_i$, which is simply its coordinates. In this case, R is 2 and $\vec{x}_i$ is (r,θ) . Similarly, a poroid (j) in class ω_p of the learning set is described by vector $\vec{x}_j^{(p)}$. The closeness of $\vec{x}_i$ to $\vec{x}_j^{(p)}$ in the pattern space is measured as the potential function, ϕ , of $\vec{x}_i^{(p)}$ at $\vec{x}_i$, $\phi(\vec{x}_i, \vec{x}_j^{(p)})$. A suitable function to use for ϕ is the

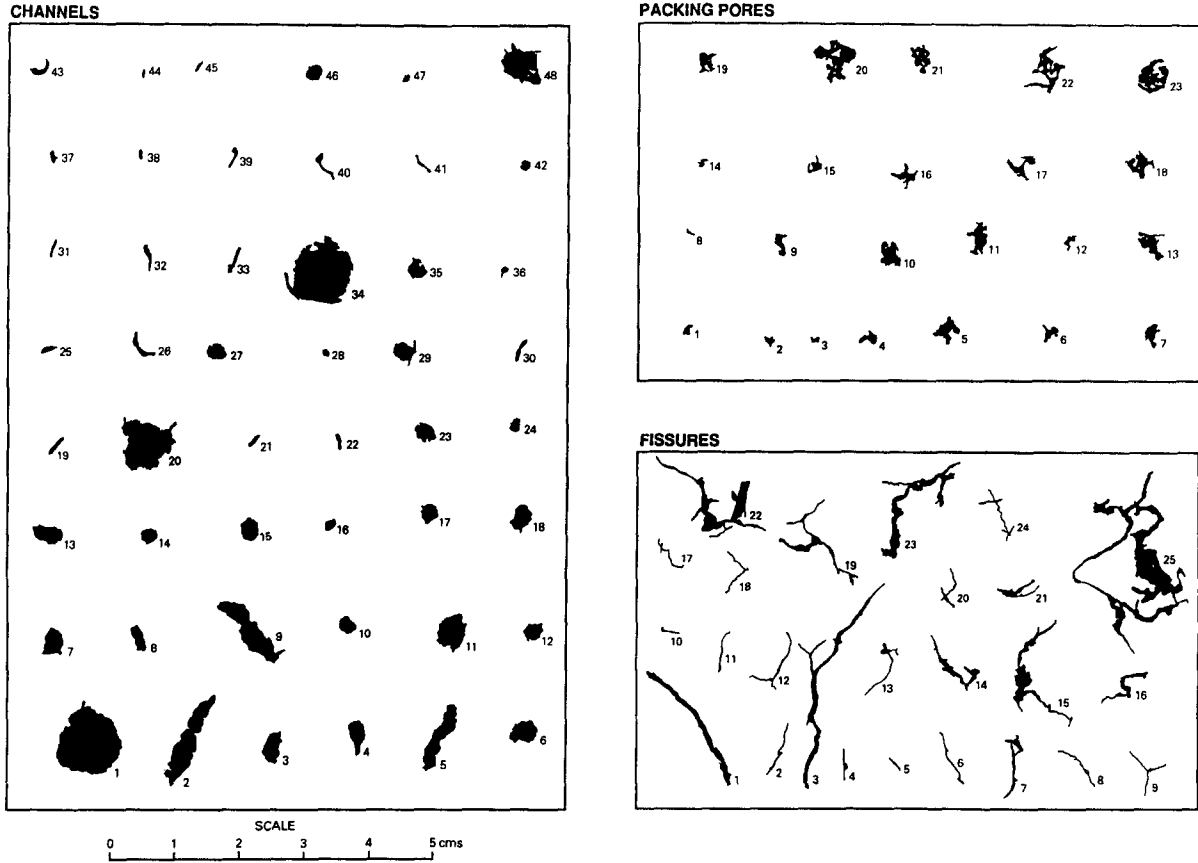


Figure 3. The poroids used as the learning set. (From Ringrose-Voase and Bullock, 1984).

Gaussian function in which case:

$$\phi(\vec{x}_i, \vec{x}_j^{(p)}) = \frac{\exp\left(-\frac{1}{2} \cdot [\vec{x}_i - \vec{x}_j^{(p)}]' \cdot \mathbf{A}_p^{-1} \cdot [\vec{x}_i - \vec{x}_j^{(p)}]\right)}{(2\pi)^{\frac{R}{2}} \cdot |\mathbf{A}_p|^{\frac{1}{2}}} \quad (7)$$

where: $\mathbf{A}_p$ = the variance matrix for class ω_p in which the covariances have been omitted for simplicity; $|\mathbf{A}_p|$ = the determinant of the variance matrix.

Because ϕ is calculated using the variances of the particular class, the value of ϕ for a given distance between $\vec{x}_j^{(p)}$ and $\vec{x}_i$ will be smaller for more tightly clustered classes. The potential functions at $\vec{x}_i$ are averaged for all the poroids of the learning set in the class to give the probability density for class σ_p at $\vec{x}_i$, $\Pr(\vec{x}_i|\omega_p)$:

$$\Pr(\vec{x}_i|\omega_p) = \frac{1}{n_p} \cdot \sum_{j=1}^{n_p} \phi(\vec{x}_i, \vec{x}_j^{(p)}) \quad (8)$$

where n_p is the number of poroids in class σ_p . This is a measure of the likelihood of a poroid having a pattern vector $\vec{x}_i$ given that it belongs to class ω_p . $\Pr(\vec{x}_i|\omega_p)$ is used to calculate $\Pr(\omega_p|\vec{x}_i)$, the probability that a poroid belongs to class ω_p given that it has a pattern vector $\vec{x}_i$, which depends on the probability densities of the other classes, ω_q . Bayes rule gives $\Pr(\omega_p|\vec{x}_i)$ as:

$$\Pr(\omega_p|\vec{x}_i) = \frac{\Pr(\omega_p) \cdot \Pr(\vec{x}_i|\omega_p)}{\sum_{q=1}^L \{\Pr(\omega_q) \cdot \Pr(\vec{x}_i|\omega_q)\}} \quad (9)$$

where L is the number of classes and $\Pr(\omega_p)$ is the a priori probability of belonging to class ω_p , which in this case is assumed to be $1/L$. $\Pr(\omega_p|\vec{x}_i)$ is calculated for each class and the poroid is allocated to the class to which it has the greatest probability of belonging.

It is also possible to calculate $\Pr(\omega_p|\vec{x}_i)$ at any point in the pattern space in order to find both the boundaries between classes and probability contours for each class (Fig. 4). ANOPOR incorporates the contours shown in Figure 4 so that a poroid can be allocated by comparing its position relative to the boundaries and probability

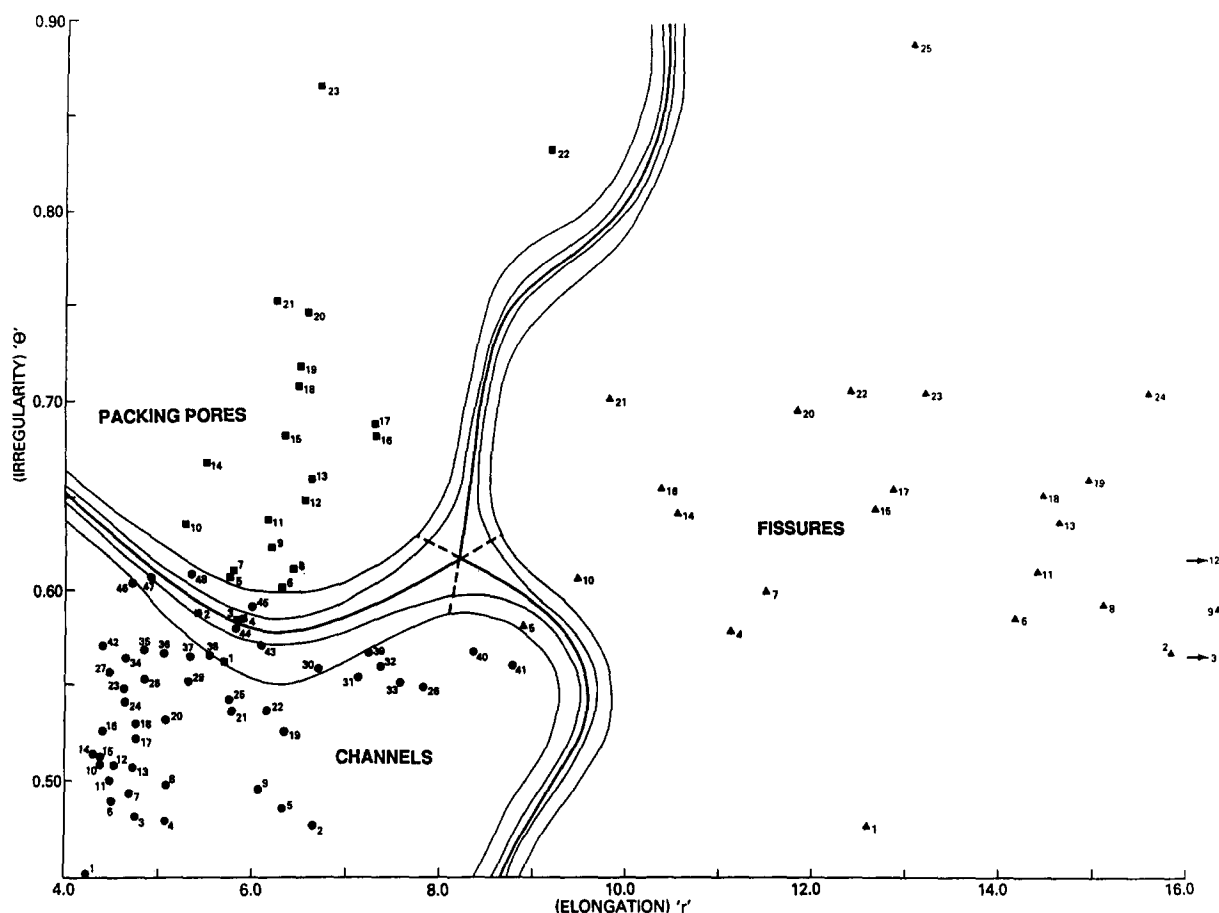


Figure 4. The learning set plotted in the r versus θ pattern space. — Class boundaries; ——— Probability contours at $p_{0.600}$ and $p_{0.525}$. $\blacktriangle$ Fissures; $\bullet$ Channels; $\blacksquare$ Packing pores. Each point refers to a poroid in Figure 3. (From Ringrose-Voase and Bullock, 1984).

contours rather than by calculating $\Pr(\omega_p | \vec{x}_i)$ each time. Poroids with a small area (< 100 pixels) are allocated directly to the most likely class. Larger poroids are allocated only if the probability is greater than 0.525 for areas between 100 and 500 pixels and 0.600 for those greater than 500 pixels. If a poroid remains unallocated, the system prompts the operator for confirmation.

Pore recognition using mathematical morphology

Mathematical morphology (Serra, 1982) offers promising techniques which might allow recognition of pore shape classes at localities within the pore space as opposed to recognition of the shape of individual poroids. These techniques can be implemented using software such as VISILOG (Noesis, S.A.R.L., Jouy en Josas, France). The ANOPOR system described above works well where there are discrete poroids in the 2-

D image, but problems are encountered where two or more classes connect and appear as a single poroid.

In 2-D, fissures appear as elongate shapes (Fig. 4). They could be distinguished using techniques from mathematical morphology as follows (Ringrose-Voase, 1989). The pore space in the binary image is reduced to its *skeleton* using the *morphological thinning* transformation described above in the section on image segmentation. To produce the skeleton, the transformation is iterated until no change occurs and all the pore space has been reduced to a line a single pixel wide (Fig. 5b). The original image is also used to produce an image of the distance transform of the pore space, in which each pixel belonging to pore space is labelled with its distance (in pixels) from the nearest boundary (i.e. solid pixel). The distance transform and the skeleton are multiplied to give the thickness of the pore space along the skeleton. For elongate objects, the contours of the distance transform tend to be parallel,

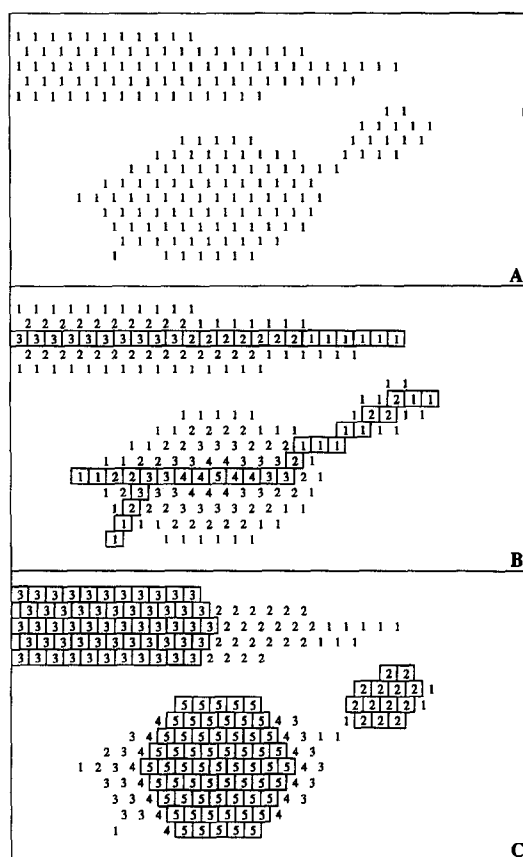


Figure 5. a. Original image digitised on an hexagonal raster showing elongate object (top) and irregular, rounded object (bottom). b. Distance transform of image. Outlined pixels show position of skeleton. c. Maximum openings (granulometry transform) of image. Outlined pixels belong to regional maxima.

whereas for rounded ones they are more concentric, as shown in Figure 5b. Hence the thickness of the pore space recorded along the skeleton changes more slowly for elongate shapes than for more rounded ones. By examining the rate of change of thickness along the skeleton, relevant parts of the pore space can be allocated to the fissure class, even if they are joined to areas of more rounded pore space.

Packing pores have more irregular outlines than channels (Fig. 4) with more non-convexities. Each non-convexity in the perimeter will initiate a branch during the skeletonisation procedure, producing a more complex skeleton. The complexity of the skeleton can be measured as the number of nodes (branches) per area of pore space or the shortness of the average length of skeleton between nodes and end-points. This measure has to be performed for a specific region of the pore space. Suitable regions can be isolated from

the pore space, irrespective of connections to other parts, by finding the *regional maxima* of the *granulometry transform*. The latter is obtained by summing the images resulting from successively larger openings of the original image. An opening of N pixels consists of an erosion of N pixels from the edge of the pore space followed by a dilation of N pixels. If the pore space at a point is less than $2N$ pixels wide, it will be deleted by the erosion, otherwise it will be eroded and then restored by the dilation. If the output of several openings are summed, the pore pixels in the summed image will be labelled with the value of N for the opening at which they were deleted. The regional maxima of the image of the granulometry transform are those plateaux surrounded only by pixels of lower value (Fig. 5c). Each regional maxima can be allocated to a pore class on the basis of the complexity of the skeleton within it. Adjacent areas of pore space can then be allocated to the same class.

Estimation of 3-D parameters using stereology

Stereology is defined by Weibel (1979) as "a body of mathematical methods relating three-dimensional parameters defining the structure to two-dimensional measurements obtainable on sections of the structure." Hence, stereology allows structural parameters of 3-D reality to be estimated from measurements made in fewer dimensions (for example, see Weibel, 1979, 1980). In order to use stereology, several conditions have to be met. First, there is an assumption that either the structure is isotropic (i.e. without preferred orientation) or the sampling is isotropic. Secondly, there must be a suitable geometric model of the structure so that *geometric probability* can be used to predict the likely outcomes of sectioning it. In general, the estimation of more complex parameters of a structure requires a more complex model. Pore type recognition allows each pore type to be treated as a separate structural component with its own model. The measurements used to estimate 3-D parameters may be obtained both manually using point counting and test lines or automatically using computerised image analysis.

Sampling strategy

Because of the reduction in the number of dimensions when sectioning a structure, unbiased sampling involves not only unbiased location of the samples but also unbiased orientation of the section. Stereologi-

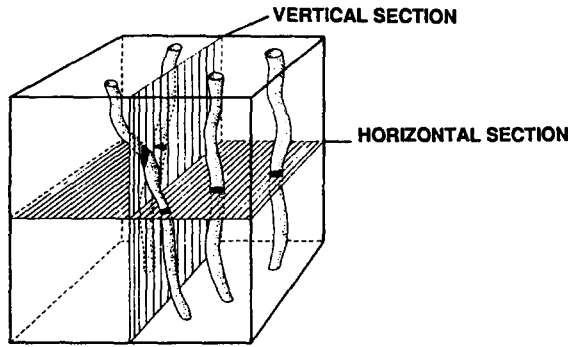


Figure 6. Channels showing preferred vertical orientation (anisotropy). (From Ringrose-Voase and Nortcliff, 1987).

cal methods assume that the structure is isotropic. If a structure is anisotropic (i.e. with preferred orientation), the effect of sectioning at different orientations can be dramatic. For example, vertically oriented channels will be intersected more frequently by horizontal than by vertical sections (Fig. 6). With such structures replicate sampling orientations should be used. To select random orientations, it is first necessary to know the probability of a particular orientation (θ , ϕ) occurring where θ is the angle to the vertical from $-\pi/2$ to $+\pi/2$ and ϕ is the horizontal angle from 0 to 2π . The probability of an angle between $\{\theta, \phi\}$ and $\{\theta + d\theta, \phi + d\phi\}$ is (Weibel, 1979):

$$\Pr(\theta, \phi) = \frac{(\sin\theta \cdot d\theta)d\phi}{2\pi} \quad (10)$$

ϕ can be selected at random. However, if random values of θ were selected, low θ values would be over-represented because the probability is proportional to $\sin\theta$. Instead, random values of $\cos\theta$ are selected so that each (θ, ϕ) combination has equal chance of being chosen.

Alternatively, θ can be selected systematically, by dividing the $\cos\theta$ range from 1 to 0 (i.e. θ from 0 to $\pi/2$) into $N/2$ equal ranges, where N is the number of orientations to be selected. The mid-point of each range (i.e. $[\cos\theta_i + \cos\theta_{i+1}]/2$) is the orientation used. The other $N/2$ orientations have corresponding, negative values. For example if $N = 6$, values of θ would be ± 0.586 ($\pm 33.6^\circ$), ± 1.05 ($\pm 60.0^\circ$) and ± 1.40 ($\pm 80.4^\circ$).

The orientation selected refers to the orientation of the normal of a 2-D section or the actual orientation of 1-D test line. In the case of the latter, Baddeley et al. (1986) point out that all orientations can be accessed by making vertical sections. The section is cut vertically

Table 1. Relationship of 3-D parameters to parameters measured in fewer dimensions. P_p is the proportion of points positioned in pore space when using point counting. Other symbols are explained in text. (After Weibel, 1979)

3-D	2-D	1-D	0-D
V_V	A_A	L_L	P_P
S_V	$\frac{4}{\pi}L_A$	$2N_L$	—
L_A	$2Q_A$	—	—

at the selected ϕ value and the test lines drawn across it at the selected θ value.

General parameters

The relationship between 3-D, 2-D and 1-D parameters is summarised in Table 1.

Volumes

Provided the correct sampling strategy has been used, various 3-D parameters can be estimated for the pore space from image measurements. The simplest of these is the volume density, V_V (volume of pore space per unit volume of soil) or macroporosity (of pore space large enough to be resolved). The volume of pore space appears as an area in 2-D when sectioned and as a length in 1-D when sampled by a test line. Hence, V_V can be estimated from the area density, A_A , or the length density, L_L .

$$V_V = A_A = L_L \quad (11)$$

This relationship is exceptional in that it does not require an assumption of isotropy (Weibel, 1979, p. 218).

Surfaces

When surfaces or sheet-like, planar structures are sectioned, they appear in 2-D as lines and in 1-D as intercepts between the surface and the test lines. The surface density, S_V (area per unit volume), can be estimated from the length density of the line that appears in 2-D, L_A , or the intercept density in 1-D, N_L .

$$S_V = \frac{4}{\pi}L_A = 2N_L \quad (12)$$

A specific example of this is the estimation of the surface area of the pore space from the perimeter density,

B_A , or the density of intercepts with pore edges, I_L .

$$S_V = \frac{4}{\pi} B_A = 2I_L \quad (13)$$

The reliability of estimates of S_V depends on the actual number of intercepts counted, N . The relative standard error, $RSE(S_V)$ is (Hillard, 1976):

$$RSE(S_V) = \frac{SE}{S_V} = \frac{1}{N^{1/2}} \quad (14)$$

where N is the number of intercepts counted along test line $L(X)$, such that $N_L = N/L(X)$. Substituting from Equation 12:

$$RSE(S_V) = \left(\frac{2}{S_V L(X)} \right)^{1/2} \quad (15)$$

Therefore the length of test line required to achieve a given $RSE(S_V)$ is:

$$L(X) = \frac{2}{S_V \cdot RSE(S_V)^2} \quad (16)$$

Lines

Structures consisting of lines in 3-D structure appear in 2-D as points where the section intersects the lines. The length density of the lines, L_V (length per unit volume) can be estimated from the intersect density, Q_A (intersects per unit area).

$$L_V = 2Q_A \quad (17)$$

Such structures cannot be sampled by test lines because the probability of a test line intercepting the lines is low.

As for surfaces, the relative standard error of L_V , $RSE(L_V)$, depends on the actual number of intersects counted, Q (Hillard, 1976):

$$RSE(L_V) = \frac{SE}{L_V} = \frac{1}{Q^{1/2}} \quad (18)$$

where Q is the number of intersects counted in an area $A(X)$, such that $Q_A = Q/A(X)$. Substituting from Equation 17:

$$RSE(L_V) = \left(\frac{2}{L_V A(X)} \right)^{1/2} \quad (19)$$

Therefore the area of image required to achieve a given $RSE(L_V)$ is:

$$A(X) = \frac{2}{L_V \cdot RSE(L_V)^2} \quad (20)$$

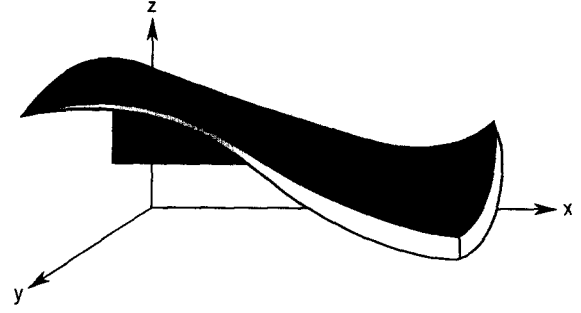


Figure 7. Planar sheet model for fissures. (From Ringrose-Voase, 1991; after Weibel, 1980).

Channel parameters

Channels can be modelled as long, gently curving cylinders of varying radii (Fig. 6) similar to the tubular model of Baddeley and Averback (1983). The volume density, V_V , and surface density, S_V , can be estimated for the channel component of the macroporosity using Equations 11 and 13 respectively.

The extent of pore space is defined here as the amount of pore space irrespective of its width (i.e. volume is proportional to extent $\times$ width). The extent of channels is the length density of their axes, L_V (length per unit volume) which can be estimated from the density of intersects with the plane of the section, Q_A , using Equation 17.

The cylindrical model allows the mean radius, $\bar{r}$, and mean square radius, $\bar{r}^2$, to be estimated from V_V , S_V and L_V (Baddeley and Averback, 1983).

$$V_V = \pi \bar{r}^2 L_V \Rightarrow \bar{r}^2 = \frac{V_V}{\pi L_V} \quad (21)$$

$$S_V = 2\pi \bar{r} L_V \Rightarrow \bar{r} = \frac{S_V}{2\pi L_V} \quad (22)$$

When a cylinder is cut by a section it gives rise to an ellipse whose minor radius is equal to the radius of the cylinder. The 3-D radius distribution can be derived by measuring the distribution of minor radii in images. The extent of channels in a radius class r_i , $L_V(r_i)$ is:

$$L_V(r_i) = 2Q_A(r_i) \quad (23)$$

where $Q_A(r_i)$ is the intersect density of channels in radius class r_i in the images.

Fissure parameters

Ringrose-Voase and Nortcliff (1987) treated fissures as sheet-like, planar structures whose width is negligible

in comparison to their extent in the other two dimensions (Fig. 7). V_V and S_V can be estimated for the fissure component of the macropore space from Equations 11 and 13 respectively. A measure of the extent of such a structure is the areal extent density $S_V(s)$. This is the area of an imaginary sheet positioned in the centre of the fissures and can be estimated in 2-D from L_A and in 1-D from N_L using Equation 12. The mean fissure width, $\bar{t}$, is:

$$\bar{t} = \frac{V_V}{S_V(s)} \quad (24)$$

However, from Equations 11 and 12 and since $L_L/N_L = \bar{t}$, the mean intercept length:

$$\bar{t} = \frac{\bar{l}}{2} \quad (25)$$

The distribution of fissure widths is most easily estimated from the distribution of the lengths of individual intercepts across the fissures. When fissures of *constant* width are intercepted by randomly oriented test lines, the length of each intercept will be greater than or equal to the 3-D width. The distribution of lengths of many intercepts can be predicted for a given width. Therefore, for fissures of varying width, the intercept length distribution resulting from each width class can be predicted. The overall intercept length distribution is the sum of that for each width class weighted according to the areal extent in that class. This allows an unknown, 3-D width distribution to be "unfolded" from a measured intercept length distribution according to a method of Cruz-Orive (1979) (see also Weibel, 1980).

The range of widths from t_0 to t_s is divided into s classes of exponentially increasing intervals. The i th class is $[t_s \Delta^{-s+i-1} < t \leq t_s \Delta^{-s+i}]$ where $\Delta = (t_s/t_0)^{1/s}$. The intercept length classes are the same but labelled j . The areal extent of fissures in class t_i , $S_V(i)$ can be estimated as follows:

$$S_V(i) = \sum_{j \leq i} \{q_{(i-j+1)} N_L(j)\} \quad (26)$$

where $N_L(j)$ is the intercept density in length class j and q_x is the x th element in a one-row matrix $\mathbf{q}_j$, which is the top line of matrix $\mathbf{P}_{ij}$. $\mathbf{P}_{ij}$ is in turn the inverse of matrix $\mathbf{P}_{ji}$ whose elements P_{ji} are calculated as follows:

$$P_{ji} = \begin{cases} |F_j - F_{j-1}|, & \text{for } j > i \\ \frac{1}{2} - F_j, & \text{for } j = i \\ 0, & \text{for } j < i \end{cases} \quad (27)$$

$$\text{where } F_j = \frac{\Delta^{2i-2j-1}}{2}$$

Continuity between fissure sheets occurs along a line (Ringrose-Voase and Nortcliff, 1987). When sectioned, this line appears in the image as the point where the outlines of fissures join. The length density of continuity, $L_v(c)$, is estimated from the intersect density of these points, $Q_A(c)$ using Equation 17. Scott et al. (1988b) define fissure connectivity density, G_V , as the number of topological loops that fissures make per unit volume. They estimated G_V manually using serial sections of area A and spacing d . A tally, G , of the disappearance or merger of loops between subsequent sections is made up to the i th section. G_V is then G/V where V is the sample volume and equal to $Ad(i-1)$.

Scott et al. (1988a) also estimated the numerical density, N_V , of connected pore networks in a clay soil using serial sections. The procedure involved manually tracking connected fissure networks through adjacent sections. N is the number of networks that disappear up to the i th section. N is estimated as N/V .

Scott et al. (1986) investigated the anisotropy of fissures by measuring N_L at different orientations in the plane of the image. In this case, the images were horizontal and information about azimuth (ϕ orientation) of the fissures was obtained. They also measured the orientation relative to the normal of the image by comparing the position of fissures in successive serial sections and calculating, in this case, the dip of the fissures, θ .

Packing pore parameters

Packing pores have a very irregular shape and it has not been possible to develop a suitable model of their geometry. Nevertheless, Equations 11 and 13 can be used to estimate their V_V and S_V respectively. When considering their extent, packing pores are best considered as a planar structure like fissures rather than as a linear one like channels. Hence, their extent is $S_V(s)$, which is estimated using Equation 12. However, the width distribution cannot be estimated using the sheet model, because their width is significant compared to their extent in the other two dimensions. The intercept length distribution can only be used to give an indication of the range of sizes. An alternative expression of the range of sizes is the opening size distribution in 2-D which is obtained using the granulometry transform described previously.

Depth functions

Moran et al. (1988), McBratney and Moran (1990) and McBratney et al. (1992) used image analysis of vertically oriented images of field-impregnated soil to derive depth functions for several parameters. The parameters were porosity, V_V , surface area density, S_V , and mean star lengths (in the horizontal plane) for pore space, $\bar{l}_p^*$, and solid, $\bar{l}_s^*$. (Mean star lengths are the mean of the length-weighted, intercept length distribution). The parameters were calculated for each row of pixels, which represented between 100 and 200 μm depth, and displayed as a function of depth. This gave information on the variation of structure with depth at previously unobtainable resolutions. Before display, smoothing splines were fitted to the raw data to remove excessive variation over small depth intervals. Moran (1990) and Moran and McBratney (1992a, b) refined this method to distinguish macropores taking part in surface infiltration from other macropores. Pores participating in infiltration were impregnated in the field with a resin containing a fluorescent dye. After excavating the sample, the remaining pores were impregnated in the laboratory using a resin with a different fluorescent dye. The different coloured fluorescence of the two dyes allowed discrimination and measurement of the two categories of pore space.

1-D and 2-D indices of macropore space

In some situations it is not possible or necessary to calculate 3-D parameters. Instead 1-D or 2-D indices can be calculated and used to compare structures of different soils or under different treatments. Ringrose-Voase (1990) discusses some 1-D parameters which can be measured using test lines. The parameters measured are the proportion of test line intercepting pore space, L_L , and the intercept density, N_L . L_L is proportional to the porosity, V_V (Equation 11), and N_L to the areal extent density, $S_V(s)$ (Equation 12). Therefore, L_L is correlated with N_L , because porosity is a function of pore extent and pore size. To use the information measured most efficiently to describe structure, it is preferable to have attributes which are independent. If $\bar{l}_p$ is the mean intercept length across pore space:

$$L_L = N_L \bar{l}_p \Rightarrow \bar{l}_p = \frac{L_L}{N_L} \quad (28)$$

The two most independent indices are then N_L and $\bar{l}_p$, which are related to the extent of the pore space and its

width, respectively. If the spacing between pore space is of concern, the mean intercept length across solid, $\bar{l}_s$, is also nearly independent of $\bar{l}_p$ if L_L is small.

$$\bar{l}_s = \frac{(1 - L_L)}{N_L} \quad (29)$$

Ringrose-Voase and Nys (1990) measured $\bar{l}_p$ and $\bar{l}_s$ for several forest soil horizons showing a variety of structures. They found $\bar{l}_s$ increases as structures become less developed (i.e. the distance between the pores defining structure increases). However, it also decreases with the size of the structural units and with the amount of intrapedal porosity.

The intercept length distributions across both pore space and solid also contain useful structural information. The information is best handled by fitting a mathematical model to the distribution so that it can be summarised in a few parameters. Scott et al. (1986) summarised the solid intercept length distribution between fissures by fitting the product of two exponentials using the Maximum Likelihood Program, MLP (Ross, 1980). The model has two parameters.

Also using MLP, Ringrose-Voase (1990) fitted a similar model consisting of the weighted sum of two exponentials, which has three parameters. This could be fitted to the distributions for both pore and solids (Fig. 8). The probability density function of this model for pore intercept lengths is:

$$f(l_p) = p \cdot a e^{-al} + (1 - p) \cdot b e^{-bl} \quad (30)$$

where $a > b$ and $0 \leq p \leq 1$. Each part of the right-hand side of the equation is an ordinary negative exponential model with parameter a or b . The two halves are weighted by p and $(1 - p)$.

This is equivalent to dividing the pore intercepts into two groups A and B. The intercept densities of each group are $N_L(A)$ and $N_L(B)$ where:

$$N_L(A) = p N_L \quad (31a)$$

$$N_L(B) = (1 - p) N_L \quad (31b)$$

Since the mean of a negative exponential of the form $f(l) = k e^{-kl}$ is $1/k$, the mean intercept lengths of the two groups are $1/a$ and $1/b$ respectively, where $1/b > 1/a$. The overall mean is:

$$\bar{l}_p = \frac{p}{a} + \frac{1 - p}{b} \quad (32)$$

Ringrose-Voase and Nys (1990) found that $1/b$ and $1/a$ indicated the range of pore sizes and p the relative frequency of smaller pores. In soils with well-developed peds, $1/b$ reflected the size of the fissures

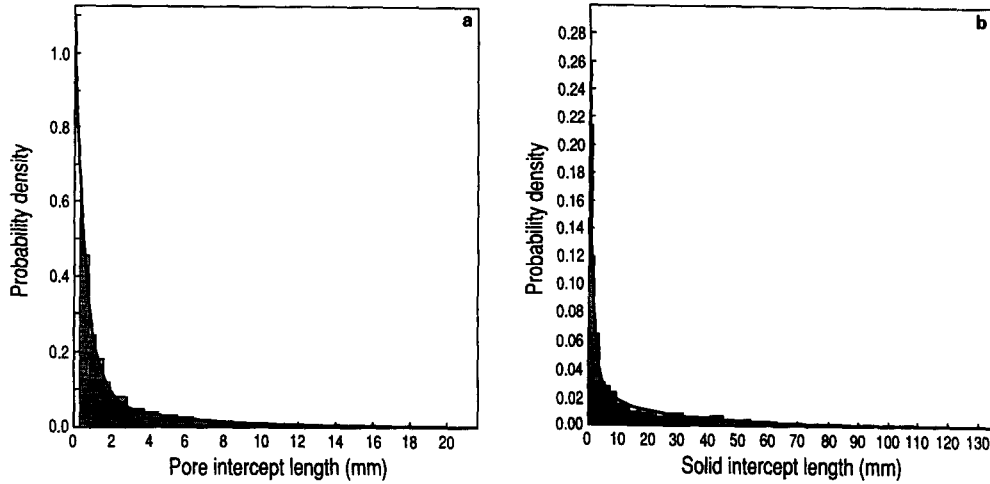


Figure 8. a. Typical pore intercept length distribution for a single image. Solid line shows the fitted model consisting of the weighted sum of two exponential probability density functions, Eq. 30. The distribution is truncated at 1.5 pixels (0.27 mm). $a = 1.722$; $b = 0.263$; $p = 0.581$; χ^2 goodness-of-fit = 29.0 on 16 degrees of freedom. (From Ringrose-Voase, 1990). b. Typical solid intercept length distribution for a single image. Solid line shows the fitted model consisting of the weighted sum of two exponential probability density functions, Eq. 33. The distribution is truncated at 0.5 pixels (0.09 mm). $c = 0.707$; $d = 0.0434$; $r = 0.379$; χ^2 goodness-of-fit = 173.7 on 17 degrees of freedom. (From Ringrose-Voase, 1990).

and $1/a$ the size of the intrapedal pores, whilst p was the relative frequency of intrapedal pores.

For solid intercept lengths, the model is:

$$f(l_s) = r \cdot c e^{-cl} + (1 - r) \cdot d e^{-dl} \quad (33)$$

where c , d and r are equivalent to a , b and p . As for the model for pore intercepts, this model divides the intercepts into two groups, C and D, with mean lengths of $1/c$ and $1/d$ respectively. The intercept densities for each group are:

$$N_L(C) = r N_L \quad (34a)$$

$$N_L(D) = (1 - r) N_L \quad (34b)$$

The proportion of the test line intercepting solid is $(1 - L_L)$. It can be shown that the proportion of this attributable to group C, $\text{Pr}(C)$ is (Ringrose-Voase, 1990):

$$\text{Pr}(C) = \frac{r/c}{(r/c) + ((1-r)/d)} \quad (35a)$$

and

$$\text{Pr}(D) = 1 - \text{Pr}(C) \quad (35b)$$

Ringrose-Voase and Nys (1990) found that for ped- al soils $1/d$, like $\bar{l}_s$, decreases with structural devel- opment and increases with the size of the structural units. However, unlike $\bar{l}_s$ it seems to be less affected by smaller, intrapedal pores. The volume of soil influ- enced by intrapedal porosity is indicated by $\text{Pr}(C)$. In

less well-developed structures, $1/d$ indicates the spac- ing between major pores, whereas $\text{Pr}(C)$ indicates the degree to which the structure is also influenced by smaller pores.

Parameters describing the solid intercept length dis- tribution show the positioning of the macropore space. For example, a large value of $1/d$ shows that there are regions of the soil that are not close to the macropore system. This might indicate that substantial volumes of the soil cannot be utilised by roots if they are using macropores as growth pathways.

Ringrose-Voase (1987) also used mean solid inter- cept length to describe structure. In this case, each part of the pore space in an image was first allocat- ed to a shape class using the statistical pattern recog- nition procedures described above. Intercept density was measured separately for channels, $N_L(C)$, fissures, $N_L(F)$, and packing pores, $N_L(P)$. Likewise area den- sity was measured for each class giving $A_A(C)$, $A_A(F)$ and $A_A(P)$, respectively. Structural units are defined by fissures and packing pores only, which together make up the structural pore space. The former define blocky and prismatic structures and the latter crumb and gran- ular structures. The intercept and area densities, $N_L(S)$ and $A_A(S)$, of this pore space are:

$$N_L(S) = N_L(F) + N_L(P) \quad (36)$$

$$A_A(S) = A_A(F) + A_A(P) \quad (37)$$

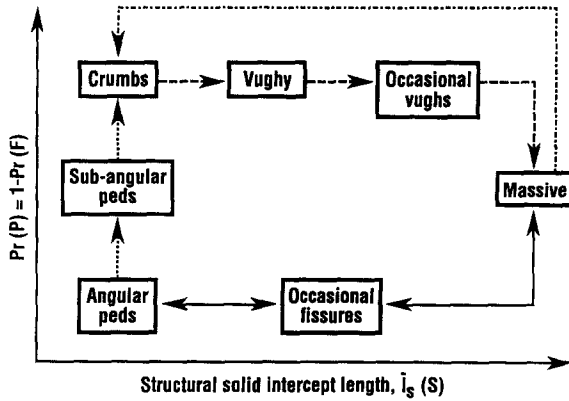


Figure 9. Positions of various structural types in the $\bar{l}_s(s)$ vs. $Pr(P)$ pattern space, showing a hypothetical sequence of structural development by various processes. — shrink/swell; - - - rounding processes; . . . settling processes; . . . tillage/selfmulching/biological activity. (From Ringrose-Voase, 1987).

Since $L_L = A_A$ (Eq. 11), the mean intercept lengths across and between structural pore space, $\bar{l}_p(s)$ and $\bar{l}_s(s)$, are:

$$\bar{l}_p(s) = \frac{A_A(s)}{N_L(s)} \quad (38)$$

$$\bar{l}_s(s) = \frac{(1 - A_A(s))}{N_L(s)} \quad (39)$$

The probabilities that an intercept with the structural pore space is with packing pores or fissures, $Pr(P)$ and $Pr(F)$, are:

$$Pr(P) = \frac{N_L(P)}{N_L(s)} \quad (40a)$$

$$Pr(F) = \frac{N_L(F)}{N_L(s)} = 1 - Pr(P) \quad (40b)$$

Ringrose-Voase (1987) plotted $\bar{l}_s(s)$ against $Pr(P)$ for soil horizons showing a range of structural types. The general pattern of the results is shown in Figure 9. Plotting different structures in this way shows relationships between them and suggests various pathways along which structural development could occur.

$\bar{l}_s(s)$ decreases with greater development of structure because the extent of structural pore space is greater. Structural development due to shrink/swell processes, especially in subsoils, creates angular blocky or prismatic structures separated by fissures and results in a high $Pr(F)$. Angular peds can be rounded by various processes to create more sub-angular peds. Such processes include slaking, differential swelling, splash erosion, freeze-thaw and biological activity. In section, parts of the pore space separating sub-angular

peds appear less like fissures and more like packing pores. Consequently $Pr(F) \approx Pr(P)$.

Rounding processes act most strongly near the surface and create more rounded peds separated by packing pores with high $Pr(P)$. Self/mulching, biological activity or tillage can also create such structures directly. As these types of structure settle due to slaking, splash erosion, compaction and shearing, packing pores become less connected and appear in section as isolated vughs. This decrease in the extent of structural pore space raises $\bar{l}_s(s)$.

Other parameters

The parameters discussed above are necessarily a limited selection. Moran (1994) presents some additional parameters which may be estimated from images of pore space. Fractals and spatial distribution are discussed briefly below because of their importance.

Fractals

Examination of the soil pore space at a range of magnifications reveals its fractal nature (Mandelbrot, 1983) because, as magnification is increased, more and finer detail in the pore space is resolved. This means that the parameter estimates discussed above are valid only for a particular magnification.

Young and Crawford (1991a) measured the fractal dimension of the surface of fractured clods of soil. They measured perimeter of the outlines using measuring units of different length. As the length of the measuring unit decreases, the perimeter increases as more detail is revealed. Hence:

$$N(r) = kr^{-D} \quad (41)$$

where r is the size of the measuring unit; N the number of units needed to measure the perimeter (i.e. $N(r)$ is the perimeter); k a constant and D the fractal dimension. In this case D is obtained from the slope of $\text{Log}(\text{perimeter})$ against $\text{Log}(r)$. In this case D was used as a measure of roughness.

Young and Crawford (1991b) calculated the fractal nature of soil aggregates by measuring the mass and mean radius manually. This gives a measure of soil heterogeneity.

Bartoli et al. (1991) used image analysis to measure the pore surface area in images of pore space at various magnifications created by transmission electron microscopy, scanning electron microscopy and optical

microscopy. The fractal nature of the pore space or solid was analysed as follows. In the centre of each image a small square of $R \times R$ pixels was analysed for the number of pore or solid pixels. A second square centred on the first but with R increased by $2 \times$ was then analysed and the number of extra pore or solid pixels calculated. This was repeated so that a total of 8 squares was analysed. The logarithm of the area of pore or solid was then plotted against the Logarithm of R . The slope of the line obtained is the fractal dimension of the pore space or solid. They found that the fractal dimension so calculated remained constant over a wide range of magnifications.

Crawford et al. (1995) measured the fractal dimension in a similar way and compared it with the slope of the moisture-release curve in a Log-Log plot. They point out that image analysis allows direct determination of the fractal properties of a soil, whereas the slope of the moisture-release curve in a Log-Log space may or may not reflect the true fractal properties of the soil.

Fractals have also been discussed in relation to linking microbial growth and structure (Crawford et al., 1993a); the number-size distributions (Crawford et al., 1993b) and hydraulic conductivity (Crawford, 1994). Kaye (1989) discusses methods of estimating the fractal dimension.

Spatial distribution

In addition to parameters concerned with the size and extent of soil pore space, it is also possible to measure the spatial distribution of pore space. Smettem and Collis-George (1985) described the spatial distribution of earthworm channels viewed macroscopically using a Poisson point process. Analysing the relative positions of features in this way can reveal whether they are randomly distributed or have a clustered or regular spacing. It can also be useful where measurement of transport properties of the soil is heavily dependent on relatively few channels occurring within the sample or measurement area. In some cases, a sample size/measurement area which includes a representative number of channels can be calculated. Where this representative size is impractical and a smaller size is used the variability of the measurement, e.g. infiltration, will be increased. It is then possible to calculate the probabilities of finding particular numbers of channels within a given area and hence the numbers of measurements needed to achieve an accurate estimate of the parameter.

Krebs et al. (1994) and Chadoeuf et al. (1994) investigated not only the spatial distribution of roots, burrows and cracks, but their distribution relative to the other features. In this way it is possible to find out whether, for example, cracks tend to occur nearer to earthworm channels than would be expected for a random distribution.

Conclusions

Computerised image analysis can produce many image parameters in 1-D or 2-D, some of which are related stereologically to 3-D parameters better than others. In any investigation, it is necessary to consider the 3-D geometry of the structure and which 3-D parameters might be of interest before deciding which 1-D or 2-D parameters are to be measured. 1-D or 2-D parameters should be chosen which allow the 3-D parameters to be estimated using stereology. In some cases, this is either not possible because of the lack of a good model or not desirable because the parameters are to be used as structural indices for comparative purposes. Even in these cases, the 1-D or 2-D parameters used should relate to the 3-D reality in a predictable manner.

Advances in technology have created possibilities for sophisticated analysis of soil macropore space using image analysis. Some applications have been described previously. To date, most of these have been studies of the effects of various treatments on soil structure using 1-D or 2-D structural indices. Such indices are especially useful when they can be measured at very small depth intervals to generate depth functions, as by Moran et al. (1988) and McBratney and Moran (1990). Others, in particular Bouma et al. (1977, 1979) and Bouma and Wösten (1984), have used data on macropore geometry to model water movement in structured soils. It is in this field that advances in the stereological estimation of 3-D parameters will be most useful.

Detailed knowledge of the geometry of pore space will also be useful in reaching a better understanding of how roots exploit the soil. Such an understanding would be enhanced if a method was found to obtain images showing both roots and pore space. Unfortunately, the dehydration and impregnation procedures described above damages many roots.

Acknowledgements

I am grateful to Dr C J Moran for many helpful discussions on this topic and for presenting this paper for

me at the "Analytical Methods for Quantifying Root and Soil Dynamics" conference at St Louis, Missouri. I also thank the conference organisers for the invitation to present this review.

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